The model: a draft

1 Indices and dimensions

	Range	
j	$1,\ldots,P$	model parameters
q	$1,\ldots,Q$	model species
r	$1,\ldots,M$	readouts
s	$1,\ldots,S$	scenarios
c	$1,\ldots,C$	study cohorts; $s(c)$ is the scenario cohort c was run under
k	$1,\ldots,K_{rc}$	statistics that cohort c reports for readout r
i	$1,\ldots,N$	simulated patients
$\mathcal{R}_c$	$\subseteq \{1,\ldots,M\}$	the readouts that cohort c reports
K_c	$\sum_{r\in\mathcal{R}_c} K_{rc}$	the rows that cohort c contributes

$\theta \in \mathbb{R}_+^P$ are the parameters and $\vartheta = \log\theta$. A *target* is a (r, c) pair. Readouts are on the log scale. n_c is the real cohort size, as from the actual data it comes from. A target names a row, not a unit of independence; the cohort is that.

2 The model

Population.

$$\vartheta_i \mid \mu, \omega \;\sim\; \mathcal{N}_P\big(\mu,\, \Omega\big), \qquad \Omega \;=\; D_\omega\, R\, D_\omega, \qquad D_\omega = \operatorname{diag}(\omega), \tag{1}$$

with R a fixed correlation matrix. Realised with common random numbers, so that ϑ_i is a smooth function of (μ, ω):

$$\vartheta_i \;=\; \mu + D_\omega L_R z_i, \qquad z_i \overset{\text{iid}}{\sim} \mathcal{N}_P(0, I_P) \;\text{ drawn once,} \qquad R = L_R L_R^\top. \tag{2}$$

Mechanism. Simulate patient i under scenario s and read off the species at the measured time:

$$y_i(s) \;=\; g\big(\vartheta_i;\, s\big) \;\in\; \mathbb{R}_+^Q. \tag{3}$$

A readout composes those species into the scalar a study reports, on the log scale:

$$x_{irc} \;=\; h_r\big(y_i(s(c))\big). \tag{4}$$

Discrepancy. Two kinds of error, distinguished by where they enter and therefore by how they propagate:

$$\tilde{x}_{irc} \;=\; \kappa_r\, h_r\Big(\underbrace{e^{\beta} \odot y_i(s(c))}_{\text{mechanism, upstream of } h_r}\Big) \;+\; \gamma_r, \qquad \underbrace{\gamma_r = Z_r^\top a, \quad \log\kappa_r = Z_r^\top b}_{\text{measurement, downstream of } h_r}. \tag{5}$$

γ and κ are the location and scale halves of a linear map on the readout.

$\beta \in \mathbb{R}^Q$ is a log-scale bias on each *species*, shared across readouts and scenarios. It enters upstream of h_r, so it propagates through the composition on its own: a ratio inherits $\beta_{q_1} - \beta_{q_2}$, a percentage inherits $\beta_q - \beta_{\text{nuc}}$, a sum inherits a weighted combination.

a and b are measurement effects on the *readout*, entering downstream of h_r, so they do not propagate. Z_r holds attributes of the readout that have no mechanistic counterpart: assay modality, and whether the quantity is a density, a fraction or a concentration.

$\beta = 0$, $a = 0$, $b = 0$ is a model with no discrepancy, that is, a model that is not biased in scale or location on any of the observed species.

The cohort vector of one patient. Stack the corrected readouts that cohort c reports, for a single patient:

$$\tilde{x}_{ic} \;=\; \big(\tilde{x}_{irc}\big)_{r\in\mathcal{R}_c} \;\in\; \mathbb{R}^{|\mathcal{R}_c|}. \tag{6}$$

Built from $\tilde{x}$ and not x: β enters inside h_r, so (5) does not factor through (4).

Cohort selection. Where cohort c reports an eligibility criterion, $w_i^{(c)} \in [0,1]$ is a fixed smooth indicator of whether patient i meets it, and $w_i^{(c)} \equiv 1$ otherwise. Write $\tilde{F}_c$ for the $w^{(c)}$-weighted empirical distribution of $\{\tilde{x}_{ic}\}_{i=1}^N$ on $\mathbb{R}^{|\mathcal{R}_c|}$, and $\tilde{F}_{rc}$ for its r-th marginal.

Statistics. Cohort c reports statistics $T_{rc1}, \ldots, T_{rcK_{rc}}$ for readout r, each a functional of a sample. The model computes the same functionals of its own predicted sample:

$$\tau_{rck}(\phi) \;=\; T_{rck}\big(\tilde{F}_{rc}\big). \tag{7}$$

Order the rows of cohort c by $r \in \mathcal{R}_c$ and then by k within r, and stack them into $\tau_c(\phi) \in \mathbb{R}^{K_c}$, and the reported values the same way into $\widehat{T}_c$. Since $\tilde{F}_{rc}$ is a marginal of $\tilde{F}_c$, the joint form changes V_c and leaves τ alone.

Observation. One cohort is one block:

$$\widehat{T}_c \;\sim\; \mathcal{N}_{K_c}\big(\tau_c(\phi),\, V_c\big), \qquad \widehat{T}_c \perp \widehat{T}_{c'} \;\text{ for } c \neq c'. \tag{8}$$

Under the ordering of (7), the diagonal blocks of V_c are the per-target covariances and the off-diagonal blocks carry the dependence between readouts of one cohort.

Sampling covariance. V_c is the covariance of the reported statistics of cohort c under resampling of its n_c patients. It has two ingredients. The first is a bootstrap from the predicted population at a plug-in ϕ_0, held fixed thereafter:

$$V_c^{\text{boot}} \;=\; \underset{*}{\mathrm{Cov}}\Big[\, T_c\big(\tilde{F}_c^*\big) \,\Big]_{\phi_0}, \qquad \tilde{F}_c^* = \text{resample of } n_c \text{ patients from } \tilde{F}_c, \tag{9}$$

where a resampled patient carries its whole vector $\tilde{x}_{ic}$ from (6). Drawing patients rather than readout values is what moves the cohort's dependence structure into V_c.

The second is E_c, from the emulator $\widehat{g}$ standing in for g. Define it on the statistics and not on the readouts, because the map from a readout error to a statistic error depends on the functional. Draw L held-out clouds $z^{(\ell)}$ at ϕ_0, evaluate each both ways, and take the covariance of the difference:

$$d_c^{(\ell)} \;=\; \tau_c^{\widehat{g}}\big(\phi_0; z^{(\ell)}\big) \;-\; \tau_c^{g}\big(\phi_0; z^{(\ell)}\big), \qquad E_c \;=\; \underset{\ell}{\mathrm{Cov}}\big[\, d_c^{(\ell)} \,\big], \qquad \ell = 1, \ldots, L. \tag{10}$$

This is $K_c \times K_c$ by construction. Report $\bar{d}_c$ beside it: a constant part of the emulator error is a fixed offset that γ_r absorbs, so only the varying part belongs in V_c.

The fallback is $E_c = \mathrm{diag}(\varepsilon^2)$, with ε_r the held-out residual of readout r repeated across its K_{rc} statistics. It asserts that the emulator errs independently across readouts, which readouts sharing a species do not satisfy, and that a readout error enters a scale statistic with the same loading as a location statistic, where the true loading is near zero.

Assembling V_c. A reported uncertainty measures $\sqrt{V_c}$ on its own diagonal, from the real patients, and carries no dependence on ϕ_0. Split the bootstrap into correlations and scales, and replace only the scales:

$$V_c \;=\; D_c\,\rho_c\,D_c \;+\; E_c, \qquad \rho_c = \mathrm{corr}\left(V_c^{\mathrm{boot}}\right), \qquad D_c = \mathrm{diag}(d_c), \qquad d_{c,rck} = \begin{cases} U_{rck}, & \text{reported,} \\ \sqrt{\left(V_c^{\mathrm{boot}}\right)_{rck,rck}}, & \text{otherwise,} \end{cases} \tag{11}$$

with U_{rck} the reported uncertainty of row (r, c, k), on the log scale. No source reports a correlation between readouts, so ρ_c always comes from the cloud. This is the split (1) already makes for the population covariance.

The ratio

$$\varrho_{rck} \;=\; U_{rck} \Big/ \sqrt{\left(V_c^{\mathrm{boot}}\right)_{rck,rck}} \tag{12}$$

is computable at ϕ_0 before anything is fitted. Above one, the predicted cloud is too narrow at that readout.

One number, one job. A standard deviation is a population spread and belongs with the scale statistics of (7). A standard error is a sampling uncertainty and belongs in D_c. At $n_c = 10$ they differ by a factor of 3.2, so the extraction has to record which one the source printed.

A source reporting only an uncertainty can supply either, not both. Send it to whichever is scarcer on that row. Where a scale statistic already exists, U goes to D_c. Where none does, U is the only statement about that width, so make it a scale statistic: the sampling standard deviation of a median at size n_c is a functional of $\tilde{F}_c$, so (7) predicts it by the bootstrap of (9) evaluated at ϕ rather than frozen at ϕ_0, and (12) becomes a residual. Freeze the resample indices alongside $z_{1:N}$ so the row stays smooth in ϕ.

Such a row carries roughly $1/\sqrt{2(n_c - 1)}$ on the log scale, the precision (14) gives a measured spread, and it is approximate because it depends on how the source built its interval. It buys precision on ω without breaking the alias: κ_r scales a sampling standard deviation exactly as it scales an interquartile range.

We may need to try to evaluate this covariance at other ϕ_0, in order to ascertain if a plug-in estimator is ok to use. Section 5 supplies the plug-in itself, at $\phi_0 = (\hat{\mu}_{\mathrm{flat}},\, \omega_0)$.

3 Priors

Location.

$$\mu \;\sim\; \mathcal{N}_P\big(\mu_0,\, \Sigma_1\big), \tag{13}$$

μ_0 and Σ_1 the mean and covariance of the stage-1 composite prior on the log scale. Recall that this is the stage 1- fitting in which we are specifically trying to learn the location parameter per QSP parameter.

Spread. Let $\mathcal{M}$ be the parameters whose between-patient spread a source has measured, with sample size n_j. For $j \in \mathcal{M}$,

$$\log \omega_j \;\sim\; \mathcal{N}\left(\log \omega_{0j},\, \frac{1}{2(n_j-1)+\tau^{-2}}\right). \tag{14}$$

For $j \notin \mathcal{M}$,

$$\log \omega_j \;=\; \log \omega_{0j} + s + u_j, \qquad s \sim \mathcal{N}(0, \tau_s^2), \qquad u_j \sim \mathcal{N}(0, \tau_\omega^2), \qquad \sum_{j \notin \mathcal{M}} u_j = 0. \tag{15}$$

Measurement discrepancy. Free, because it has no parameter twin. Z has an intercept, so a_1 and b_1 are the global location and width corrections.

$$a \;\sim\; \mathcal{N}(0,\, \sigma_a^2 I), \qquad b \;\sim\; \mathcal{N}(0,\, \sigma_b^2 I), \qquad \sigma_a = \sigma_b = 0.5. \tag{16}$$

Mechanism discrepancy. Tightly shrunk, because it competes with μ:

$$\beta_q \;=\; \tau_\beta\, \beta_q^{\text{raw}}, \qquad \beta_q^{\text{raw}} \sim \mathcal{N}(0,1), \qquad \log \tau_\beta \sim \mathcal{N}\big(\log 0.15,\, 0.5^2\big). \tag{17}$$

β_q is set to zero for every species outside a declared subset $\mathcal{S} \subset \{1, \ldots, Q\}$. The default $\mathcal{S}$ is the shared denominators, nucleated_total and V_T.

4 Posterior and deliverable

$$\phi \;=\; \big(\mu,\, s,\, u^{\text{raw}},\, \log \omega_{\mathcal{M}},\, a,\, b,\, \beta_{\mathcal{S}}^{\text{raw}},\, \log \tau_\beta\big). \tag{18}$$

$$p(\phi \mid \mathcal{D}) \;\propto\; p(\phi) \prod_{c=1}^{C} \mathcal{N}\big(\widehat{T}_c;\, \tau_c(\phi),\, V_c\big). \tag{19}$$

The virtual population is the posterior predictive over patients,

$$p(\theta_{\text{new}} \mid \mathcal{D}) \;=\; \int \mathcal{N}_P\big(\log \theta_{\text{new}};\, \mu,\, \Omega\big)\, p(\phi \mid \mathcal{D})\, d\phi, \tag{20}$$

and it is emitted as an ensemble: one population per posterior draw of ϕ, so that between-patient variability and uncertainty about ϕ stay separable in whatever is computed downstream.

5 The flat fit

The flat fit estimates the centre of the population and holds its shape fixed. Set $\omega = \omega_0$ and drop the three spread terms from the parameter vector:

$$\phi_{\text{flat}} \;=\; \big(\mu,\, a,\, b,\, \beta_{\mathcal{S}}^{\text{raw}},\, \log \tau_\beta\big). \tag{21}$$

Nothing else changes. Equations (2) to (19) run as written, with ω a constant rather than a function of ϕ.

Why ω_0 and not zero. Setting $\omega = 0$ collapses the cloud onto one point, and three things die with it. $\tilde{F}_c$ becomes a point mass, so (9) returns zero and V_c has to come from outside the model. $w^{(c)}$ becomes a switch on the whole cohort instead of on patients, so the eligibility rule stops working and turns discontinuous in μ. And every location functional of a point mass returns the same number, so a cohort that reports a mean beside a median predicts one value for two rows.

Each of those is a diagnostic, and diagnostics are what the flat fit is for. The third is the sharpest: the gap between a reported mean and a reported median is the skewness of the pushforward, which at $\omega = \omega_0$ the model predicts rather than absorbs.

What it buys.

V_c is computable	(9) resamples a real cloud, so the bootstrap works unchanged.
The scale rows are held out	Predict them at ω_0 and report the ratio per readout. That is the width shortfall, before any population parameter is estimated.
$\hat{\mu}$ needs no translation	It is already a population centre, so there is no Jensen gap to correct afterwards.
It supplies the plug-in	Take $\phi_0 = (\hat{\mu}_{\text{flat}}, \omega_0)$ for (9).

Fit the location rows and hold the scale rows out, so the ratio above is an honest comparison. Refitting with them included is the natural second run, and it is what identifies b.

Cost. A gradient step still costs $C \times N$ forward passes, but N only has to resolve the location statistics and a rough width, not the flanks at a precision that supports inference on ω, so it sits far below the N of the population fit.

What it cannot test. The shape of the population: ω, R, and the direction s and b_1 share. It does test the prior on μ, the forward map, the discrepancy terms, the block structure of (8) and the sampler, and a fault in any of those is one the population fit would also have.

6 Fixed inputs

R	Population correlation matrix. Not identified by these data.
ω_0	Prior centre of the spreads, per parameter.
μ_0, Σ_1	Stage-1 composite prior, log scale.
h_r	Readout composition, including unit conversions. Already in the target definitions.
Z	Readout attributes for the measurement discrepancy: assay modality, quantity type.
$\mathcal{S}$	Species allowed a mechanism discrepancy. Default: the shared denominators.
$z_{1:N}$	The latent cloud, drawn once.
$w^{(c)}$	Eligibility weights, read from each paper's criterion.
V_c	Assembled by (11) at ϕ_0, then frozen. One block per cohort.
E_c	Emulator error covariance on the statistics of cohort c, from (10).
U_{rck}	Reported uncertainty of a statistic, where a source gives one. Log scale.
τ_s, τ_ω, τ, σ_a, σ_b	Prior widths.

7 Known simplifications

- **Independence across cohorts.** Equation (8) is block diagonal by cohort, so what remains asserted is that distinct cohorts are distinct patients. That holds across studies. It fails where two cohorts come from one publication and share people, as when a subgroup is reported beside the whole. Check it before splitting any pooled target: a split that manufactures two cohorts from one set of patients moves the problem instead of removing it.

Report the effective row count $K_c/\{1 + (K_c - 1)\bar{\rho}_c\}$ at mean off-diagonal correlation $\bar{\rho}_c$. The source supplying 19 of the 49 targets, from about ten patients per arm, is the one to look at first.

- **No study offset.** A free η_c would be aliased with the measurement discrepancy. It is omitted and γ_r absorbs the study effect, so γ_r mixes assay bias with study bias.
- V **frozen at** ϕ_0**.** Equation (19) is an estimating equation, not a likelihood. Letting V move with ϕ would reintroduce a $-\frac{1}{2}\log\det V$ term that rewards a narrow population regardless of fit.
- s **and** b_1 **share a direction.** To first order every scale statistic depends on the global width multiplier and the global width correction only through their sum, so the split between a narrow simulator and a narrow population is decided by τ_s and σ_b unless some readout is driven mainly by $\mathcal{M}$. Report the joint posterior of the pair.
- Z **is asserted, not tested.** The claim that assay modality and quantity type have no mechanistic twin, while species-level bias does, is an argument rather than a result. The test is to project each column of Z onto $\mathrm{col}\big(\partial\tau/\partial(\mu,\omega)\big)$ and read the residual norm. That needs the emulator, and it has not been run.